Homework 4

ECON 470, Spring 2026

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1. Hospitals Filing Multiple Reports in the Same Year

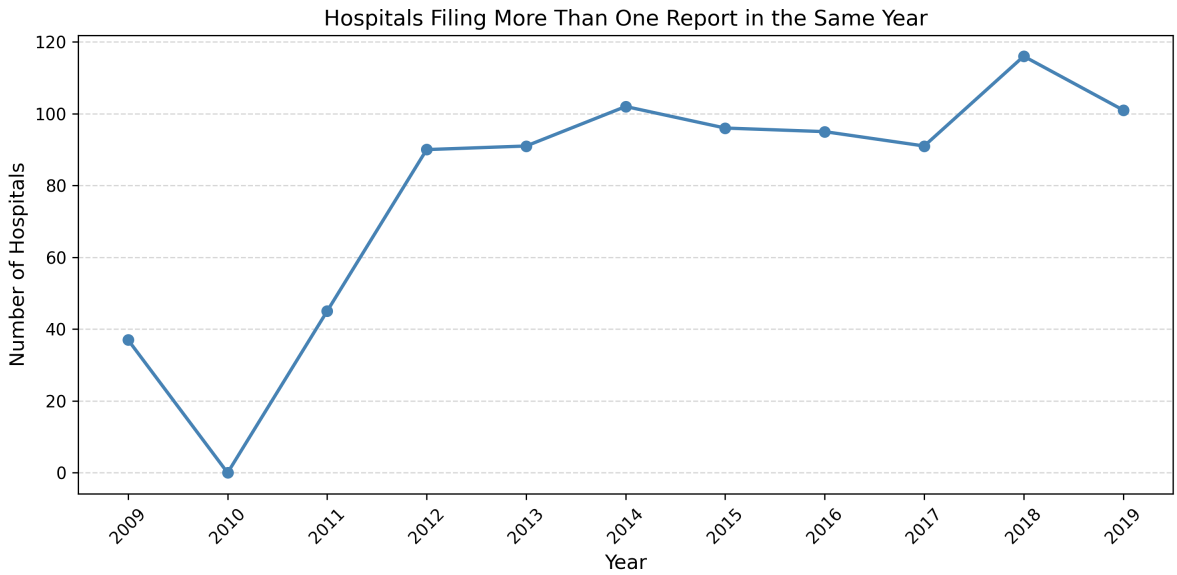

Figure 1: Number of hospitals filing more than one report per year, 2009–2019

Q1 answer: The number of hospitals filing duplicate reports ranges from 0 in 2010 to a peak of 116 in 2018. This typically happens when hospitals change fiscal years and submit two reports covering the same calendar year. Resolving duplicates before analysis is essential to avoid double-counting charges, payments, and discharges.

2. Unique Hospital IDs After Removing Duplicates

Q2 answer: After collapsing to one report per hospital per fiscal year — preferring final over late over not-yet-submitted reports and taking the most recently submitted record when ties remain — there are **6,850** unique Medicare provider numbers in the data across 2009–2019.

3. Distribution of Total Charges by Year

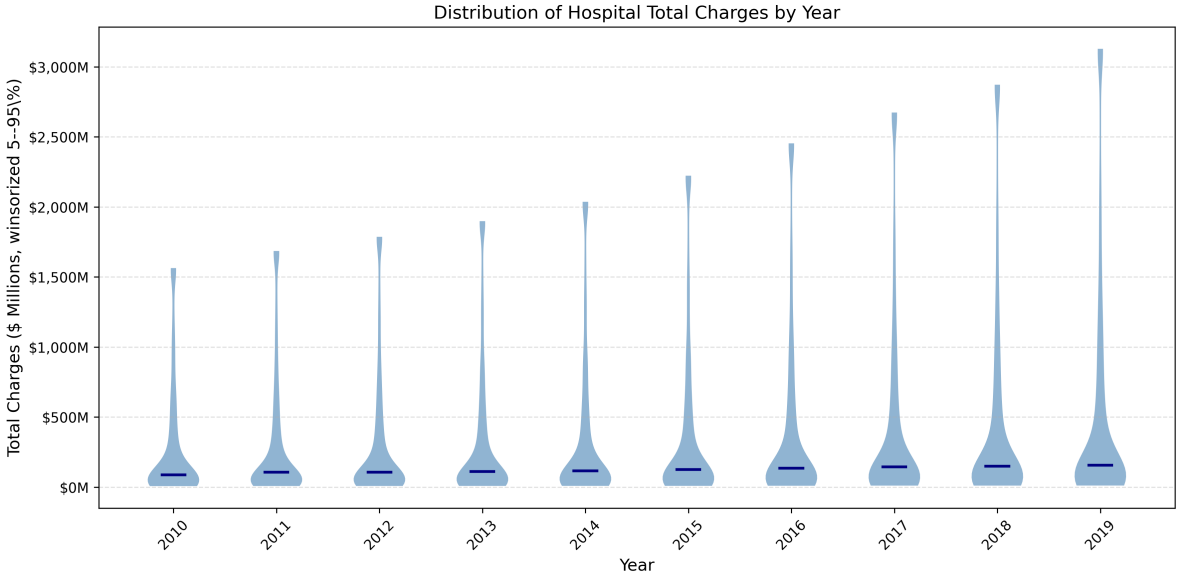

Figure 2: Distribution of total charges by year, winsorized at 5th and 95th percentiles

Q3 answer: Total charges rise steadily from 2010 to 2019, with both the median and upper tail increasing each year. The distribution is right-skewed in every year, driven by a small number of large academic hospitals with very high charges.

4. Distribution of Estimated Prices by Year

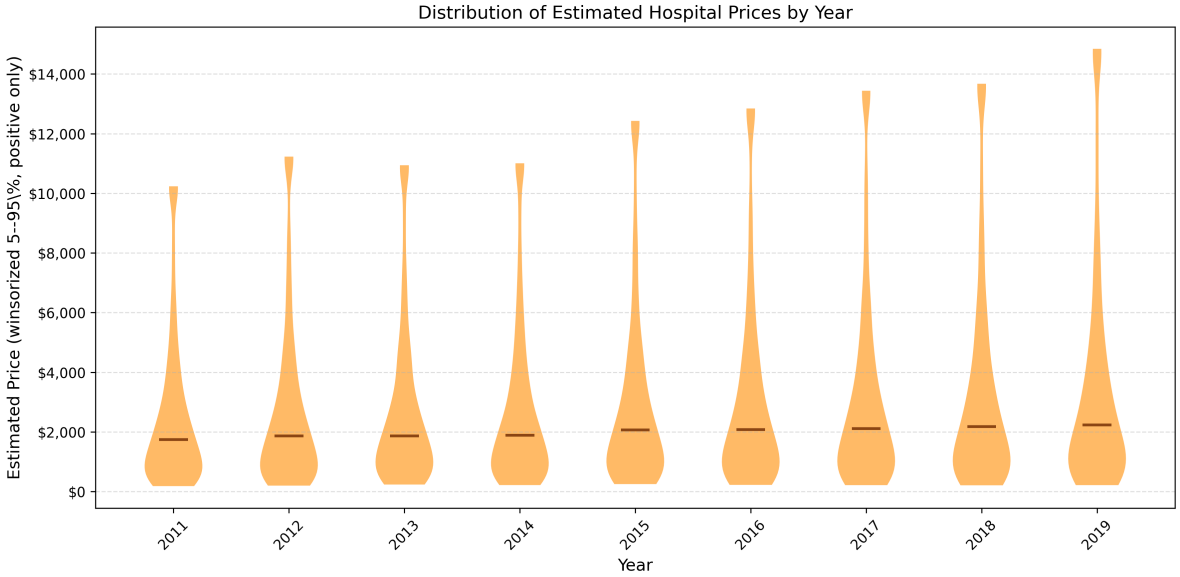

Figure 3: Distribution of estimated prices by year, winsorized at 5th and 95th percentiles, positive values only

Q4 answer: Estimated prices drift upward from 2010 to 2019, consistent with hospitals gaining leverage in commercial negotiations over time. The distributions are more symmetric than total charges since this measure nets out Medicare payments and scales by non-Medicare discharges.

5. Share of Hospitals Penalized Under HRRP and/or VBP

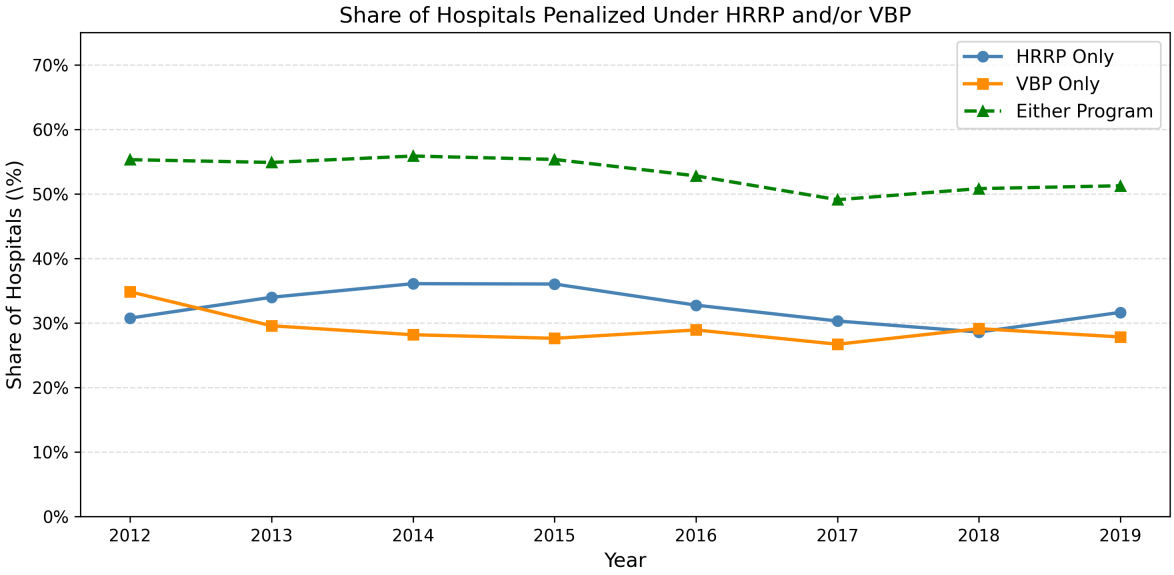

Figure 4: Share of hospitals penalized under HRRP and/or VBP, 2012–2019

Q5 answer: Roughly half of hospitals are penalized under at least one program each year. HRRP penalization peaks around 2014–2015 then declines as hospitals adapted, while VBP stays relatively flat since it mechanically penalizes roughly half of participants by design.

6. OLS Estimates of Net Penalties on Price Changes

Table 1: OLS estimates of the effect of net penalties on price changes

	Spec (1)	Spec (2)	Spec (3)
Net Penalty ($1000s)	0.145	0.127	0.121
	(0.037)	(0.040)	(0.040)
Avg Beds (pre-2012)		1.137	-0.143
		(0.614)	(0.786)
Avg Medicaid Discharges			0.013
			(0.007)
N	2,868	2,868	2,868
R^2	0.014	0.016	0.019

7. Net Penalty vs Pre-2012 Medicare Discharges

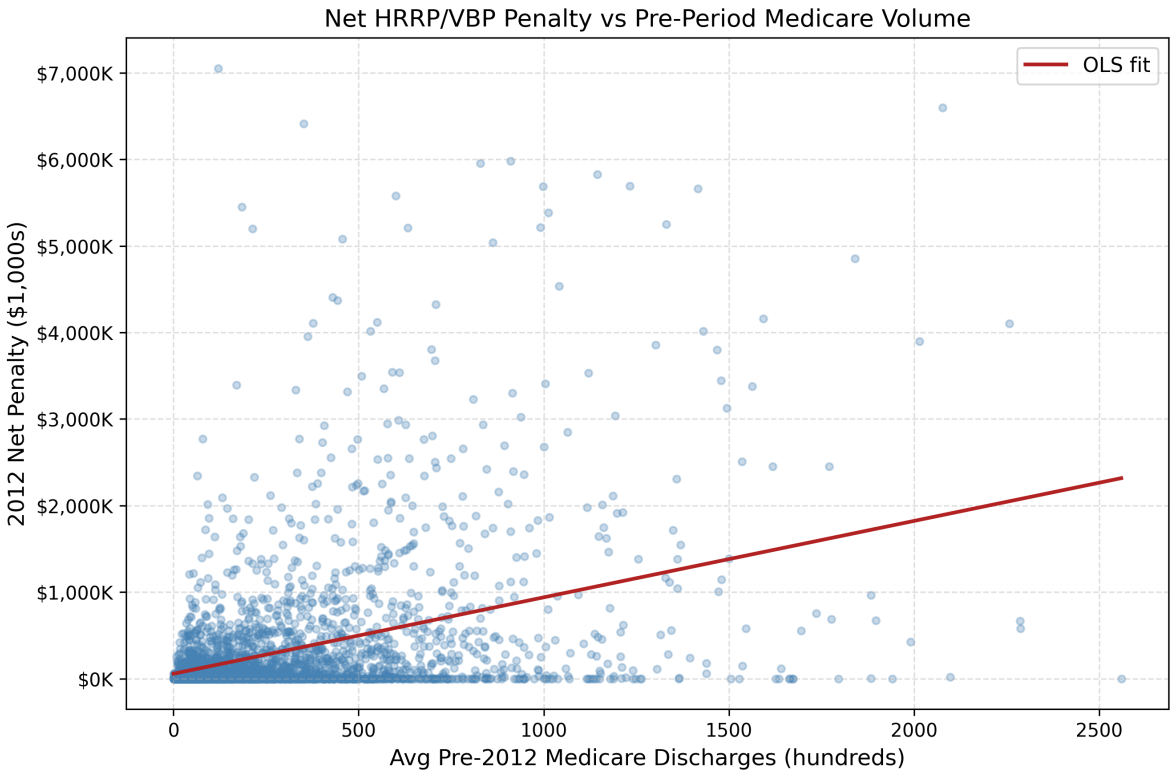

Figure 5: Net HRRP/VBP penalty vs average pre-2012 Medicare discharges (hundreds)

8. First Stage and Reduced Form

Table 2: First stage — net penalty on pre-2012 Medicare discharges

	Spec (1)	Spec (2)	Spec (3)
Avg Medicare Discharges (100s)	1.664	2.754	2.670
	(0.198)	(0.549)	(0.565)
Avg Beds (pre-2012)		-5.968	-6.133
		(2.302)	(2.330)
Avg Medicaid Discharges			0.006
			(0.008)
N	2,868	2,868	2,868
R^2	0.158	0.169	0.169
F-stat	70.4	49.3	32.8

Table 3: Reduced form — price change on pre-2012 Medicare discharges

	Spec (1)	Spec (2)	Spec (3)
Avg Medicare Discharges (100s)	0.328	-0.315	-0.599
	(0.116)	(0.228)	(0.251)
Avg Beds (pre-2012)		3.516	2.955
		(1.067)	(1.096)
Avg Medicaid Discharges			0.019
			(0.008)
N	2,868	2,868	2,868
R^2	0.004	0.007	0.012

9. IV Estimates of Net Penalties on Price Changes

Table 4: OLS vs IV estimates of net penalties on price changes

	Spec (1)	Spec (2)	Spec (3)
OLS: Net Penalty ($1000s)	0.145	0.127	0.121
	(0.037)	(0.040)	(0.040)
IV: Net Penalty ($1000s)	0.197	-0.114	-0.224
	(0.074)	(0.089)	(0.114)
OLS: Avg Beds (pre-2012)		1.137	-0.143
		(0.614)	(0.786)
IV: Avg Beds (pre-2012)		2.834	1.579
		(0.711)	(0.850)
OLS: Avg Medicaid Discharges			0.013
			(0.007)
IV: Avg Medicaid Discharges			0.020
			(0.008)
N (OLS)	2,868	2,868	2,868
N (IV)	2,868	2,868	2,868

10. Local Average Treatment Effect

Q10 answer: The IV estimates identify a Local Average Treatment Effect (LATE) for compliers — hospitals whose penalty status was determined by their pre-2012 Medicare volume. This is likely mid-sized hospitals at the margin of penalty exposure. Very small hospitals with too few Medicare patients to be penalized regardless, and very large academic centers penalized regardless of volume, are not identified by the instrument. The LATE could differ from the overall ATE if large safety-net hospitals respond more aggressively to penalties than the mid-sized compliers our instrument captures.